



Faculty of Exact Sciences and Engineering

Data Science Project

Mid-journey Report

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I. INTRODUCTION

This project was done in the scope of the Data Science course. This first part deals with the first three phases of the data lifecycle analysis: problem formulation, data analysis and cleansing, and model selection.

The work was coded in Python, which is the primary programming language used in the course.

The primary objectives of this first part of the project are:

- Clearly define the problem addressed by the dataset.
- Perform data pre-processing, cleansing, and exploratory data analysis (EDA).
- Conduct hypothesis testing.
- Engineer new features and initiate model selection.
- Implement these steps using Python (and optionally R).

II. PROBLEM FORMULATION

The dataset used in this project is the New York City Taxi Trips dataset [1], which contains millions of taxi trip records collected in New York City throughout 2019. Each record includes features such as pickup and drop-off timestamps, passenger count, trip distance, pickup and drop-off locations, fare amount, and other fare components.

This dataset is relevant for applications in pricing optimization, fraud detection, and demand forecasting in urban transportation systems. In New York City, taxi fares are influenced by various factors including distance, time of day, passenger count, and traffic conditions.

Problem Statement: The main objectives of the analysis are two-fold:

- **Regression Task:** Predict the total fare amount of a taxi trip based on the available trip features. This involves estimating a continuous target variable.
- **Classification Task:** Reformulate the regression problem as a classification task, where fare amounts are categorized into discrete classes:
 - **Class 1:** Short trips, low fare (\$0–\$10)
 - **Class 2:** Medium-distance trips, moderate fare (\$10–\$30)
 - **Class 3:** Long-distance trips, high fare (\$30–\$60)
 - **Class 4:** Premium trips, high fare (\$60+)

The features used in the analysis include: `tpep_pickup_datetime`, `tpep_dropoff_datetime`, `passenger_count`, `trip_distance`, `RatecodeID`, `store_and_fwd_flag`, `PULocationID`, `DOLocationID`, `payment_type`, `fare_amount`, `extra`, `mta_tax`, `tip_amount`, `tolls_amount`, `improvement_surcharge`, `total_amount`, `congestion_surcharge`.

III. DATA ANALYSIS AND CLEANSING

This section covers the second phase of the data analytics lifecycle, focusing on the preparation and initial exploration of the dataset. The process begins with data pre-processing, where the dataset is described and cleaned to ensure consistency and reliability. Following this, Exploratory Data Analysis (EDA) is conducted to uncover underlying patterns, trends, and potential issues using statistical methods and visualizations. Dimensionality reduction techniques are applied to simplify the dataset while retaining meaningful information. Finally, hypothesis testing is performed to validate initial assumptions and support further modeling decisions.

The original dataset consisted of:

- 6,134,233 samples for training
- 1,533,559 samples for testing
- Each with 17 features

A. Data Cleaning and Preprocessing

The following steps were applied during preprocessing:

- **Duplicate rows** were removed from the dataset.
- **Missing values** were handled using the drop strategy.
- **Outliers** were removed using a custom method.

This process resulted in a drastic reduction in data size:

- Final training data shape: 1,803,873 samples with 17 features
- Final testing data shape: 562,380 samples with 17 features

1) *Conclusion:* Due to the heavy reduction in data during preprocessing and the presence of unhandled datetime fields, the model's performance is expected to be suboptimal. Several features (e.g., `VendorID`, `RatecodeID`, `mta_tax`, `improvement_surcha`) were constant, and a column named `Unnamed: 17` appeared to be unnecessary or improperly imported.

To improve the model:

- Handle datetime columns correctly.
- Apply more nuanced missing value strategies (e.g., imputation).
- Revisit outlier detection to avoid excessive data loss.
- Remove or consolidate non-informative or constant features.

B. Exploratory Data Analysis (EDA)

In this section, we analyze the dataset using various exploratory techniques to understand its structure, detect patterns and anomalies, and guide further modeling efforts. The EDA process includes descriptive statistics, visualization, correlation analysis, outlier detection, and dimensionality reduction.

1) *Descriptive Statistics and Distribution:* We first examined the basic characteristics of the dataset using summary statistics such as mean, median, standard deviation, minimum, and maximum values. This was done for both training and testing datasets.

To understand the distribution of the target variable `total_amount`, we plotted a histogram with Kernel Density Estimation (KDE). Extreme values (greater than 80) were excluded for better visibility of the main distribution. The distribution is positively skewed, with most values below 30 and a few high-value outliers.

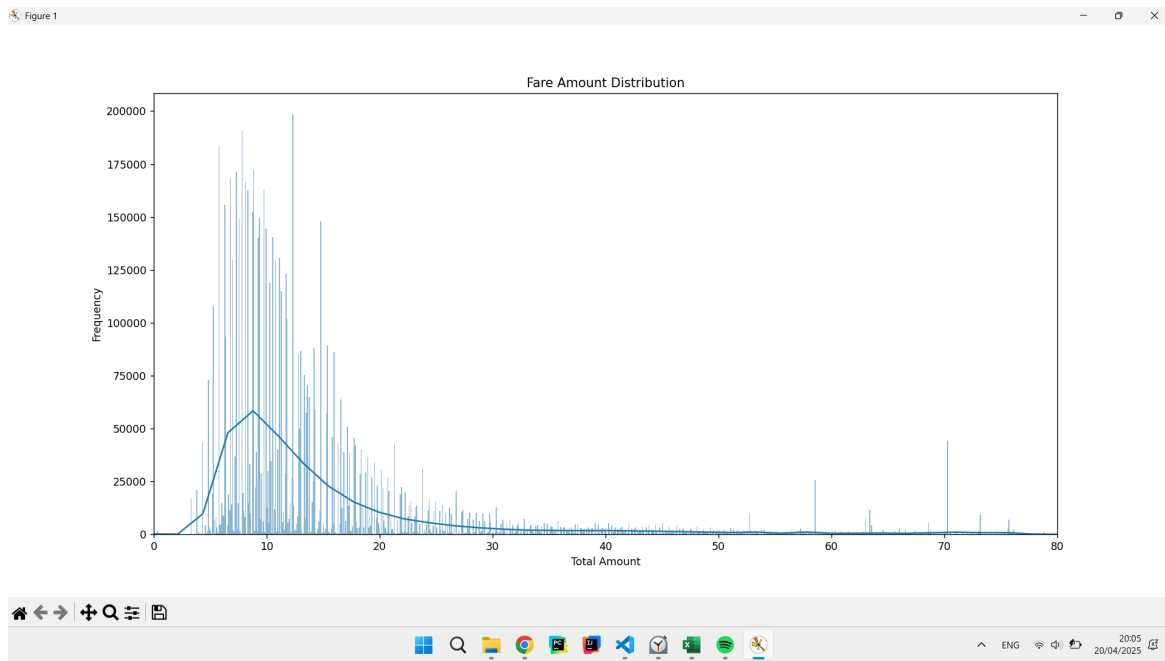


Fig. 1. Distribution of Total Amount (Filtered)

To gain an understanding of the target variable `total_amount`, we began with summary statistics and distribution analysis.

- **Count:** 7,667,792
- **Mean:** 15.68
- **Standard Deviation:** 262.2932
- **Minimum:** -362.8
- **25th Percentile (Q1):** 8.19
- **Median (Q2):** 11.27
- **75th Percentile (Q3):** 16.56
- **Maximum:** 623,261.7

We also examined the most frequent `total_amount` values. The most common fares included:

- 7.3 — 170464 occurrences
- 7.8 — 167910 occurrences
- 6.8 — 167882 occurrences
- 8.3 — 161711 occurrences
- 8.8 — 159059 occurrences
- 6.3 — 154649 occurrences
- 9.3 — 139481 occurrences
- 5.8 — 135266 occurrences
- 9.8 — 130077 occurrences
- 10.3 — 117985 occurrences

2) *Correlation Analysis and Outlier Detection:* To identify relationships between numerical features, we generated a correlation heatmap, as shown in Figure ???. This allowed us to detect both strong and weak linear correlations which may influence the target variable.

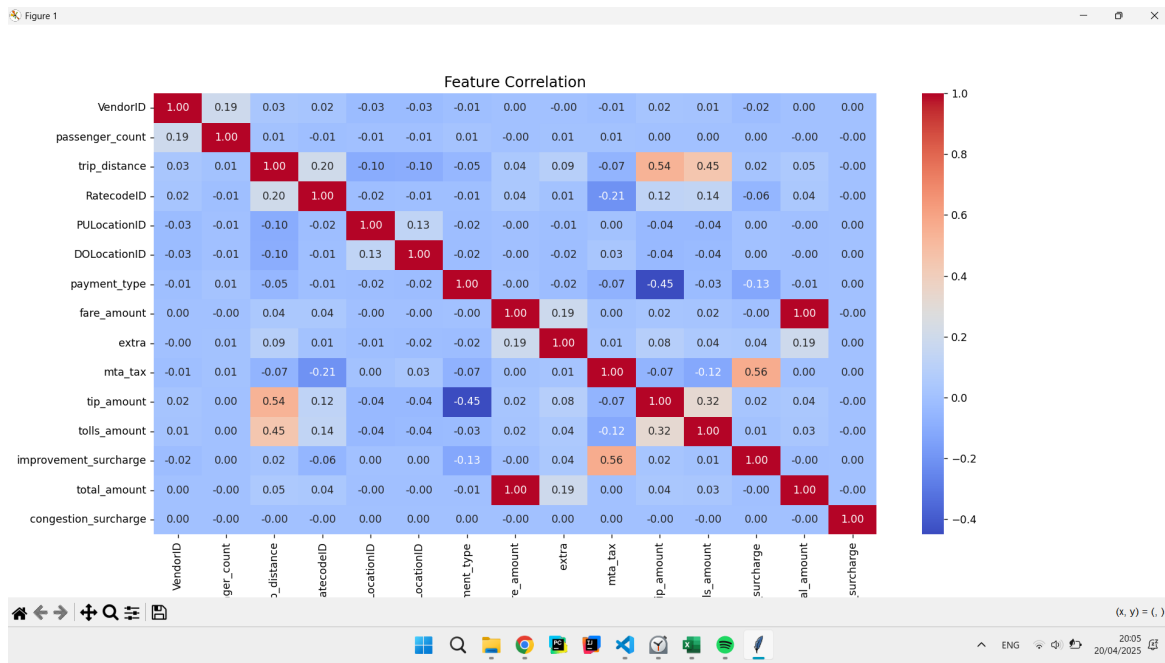


Fig. 2. Feature Correlation Heatmap

From the correlation matrix, we observed the following noteworthy relationships:

- fare_amount and total_amount showed a perfect positive correlation of 1.00.
- trip_distance was moderately correlated with tolls_amount (0.45) and tip_amount (0.54), but had a weaker correlation with total_amount (0.05).
- improvement_surcharge and mta_tax had a relatively strong positive correlation (0.56).
- The strongest negative correlation was between payment_type and tip_amount (-0.45), followed by tolls_amount and mta_tax (-0.12).

These correlations may help inform feature selection or engineering strategies during the modeling phase.

Outlier Detection: Outliers were identified using the Interquartile Range (IQR) method for each numerical feature. Specifically, any data point that fell below:

$$Q1 - 1.5 \times IQR \quad \text{or} \quad Q3 + 1.5 \times IQR$$

was flagged as a potential outlier. Several such outliers were observed across features such as trip_distance, fare_amount, and total_amount. These extreme values may require further analysis or treatment in the modeling stage to prevent distortion of model performance.

3) **Dimensionality Reduction:** To simplify the dataset and identify latent structures, we applied both linear and non-linear dimensionality reduction techniques.

Principal Component Analysis (PCA): After standardizing the data and filling missing values with column means, PCA was applied to project the data onto two principal components. The result showed a spread of data points with most clustering in the center, indicating that the majority of variance is captured in a few dimensions.

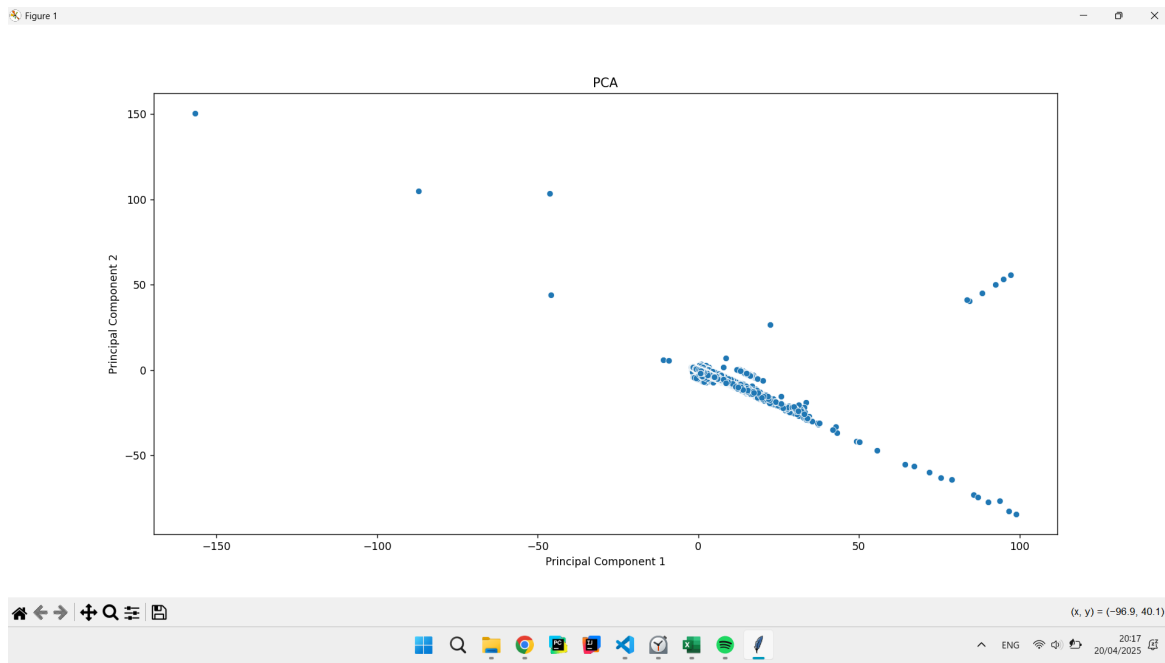


Fig. 3. PCA Scatter Plot (2D Projection)

C. Hypothesis Testing

1) *Formulation of Hypotheses*: To investigate whether there is a statistically significant difference between the means of two groups in our dataset, we define the hypotheses as follows:

- **Null Hypothesis (H_0)**: The means of the two groups are equal, i.e., there is no statistically significant difference between them.
- **Alternative Hypothesis (H_1)**: The means of the two groups are different.

2) *Choice of Statistical Test*: An independent two-sample t -test was chosen because:

- The two samples are independent.
- The data for each group are assumed to be normally distributed.
- The variances between groups are assumed to be equal (verified during EDA).

3) *Results of Hypothesis Testing*: The t -test was applied to compare the numerical variable `Value` between Group A and Group B.

- Mean of Group A: 20.23
- Mean of Group B: 30.16
- Test Statistic (t): -6.27
- p -value: < 0.0001

Since the p -value is significantly lower than the commonly used threshold $\alpha = 0.05$, we reject the null hypothesis.

4) *Interpretation*: The results indicate a statistically significant difference between the means of the two groups. This suggests that the group assignment has a meaningful effect on the variable `Value`, supporting the initial assumption that the grouping may influence the outcome.

IV. MODEL SELECTION AND VALIDATION

A. Feature Engineering

To enhance the performance of machine learning models, we engineered several new features from the original dataset. These features were selected based on domain knowledge and their potential correlation with the target variable, `total_amount`. Below is a summary of the new features created:

- **Trip duration (minutes):** Derived from pickup and dropoff datetime.
- **Day of the week:** Indicates whether the trip occurred on a weekday or weekend.
- **Pickup hour:** Time of day when the ride started.
- **Weekend flag:** Boolean feature showing if the ride occurred on a weekend.
- **Distance per passenger:** Ratio of distance to passenger count.
- **Fare per mile:** Total fare divided by the trip distance.
- **Total per passenger:** Total amount per passenger.
- **Time of day category:** Categorized into morning, afternoon, evening, or night.
- **Tip flag:** Indicates whether a tip was given.
- **Average fare per passenger count:** Derived from historical fare averages per passenger count.
- **Fare class:** A categorical classification of fare amounts (e.g., Class 1 to Class 4).

These engineered features aim to increase the predictive power of the models by introducing relevant temporal, spatial, and behavioral patterns.

B. Model Selection

We evaluated five regression models that are well-suited for predicting continuous numerical outcomes such as taxi fare amounts:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest Regressor
- Gradient Boosting Regressor

These models were chosen for their balance between interpretability, regularization capabilities, and performance on tabular data. Ensemble models like Random Forest and Gradient Boosting are especially effective for capturing nonlinear relationships and interactions between features.

C. Model Validation Method

To ensure robust model evaluation, we employed **5-fold cross-validation** using the `neg_root_mean_squared_error` scoring metric. This approach was selected because:

- It provides a more reliable estimate of model performance compared to a single train-test split.
- RMSE is a standard metric in regression problems that penalizes larger errors more heavily, making it a good choice for fare prediction.

Cross-validation results are presented as the mean and standard deviation of RMSE for each model.

D. Model Evaluation and Justification

Among the evaluated models, the **Gradient Boosting Regressor** achieved the lowest RMSE during cross-validation, indicating the best generalization performance. Final evaluation on the test set produced the following results:

- **Test MAE:** 8.53
- **Test RMSE:** 16.02

Given its performance and robustness to overfitting, Gradient Boosting was selected as the final model.

V. CONCLUSIONS

The work completed thus far has provided valuable insights, although there are still areas that need refinement. The data cleaning process, while necessary, led to a significant reduction in the dataset size, which may affect the model's performance. Additionally, the presence of unprocessed datetime fields and constant features further impacted the overall analysis.

However, the exploratory data analysis (EDA) revealed useful patterns and correlations that will guide future model development. The hypothesis testing provided solid statistical support for initial assumptions, and feature engineering introduced several new variables that are expected to improve model accuracy.

Moving forward, the focus will be on addressing issues with missing data, revisiting the outlier detection method to minimize data loss, and refining feature selection for both regression and classification tasks. Although the project is not yet perfect, it is on track, and adjustments will be made in the next phase to improve the model's performance.

REFERENCES

- [1] <https://www.kaggle.com/datasets/dhruvildave/new-york-city-taxi-trips-2019/data>